



DesignWare® Building Block IP

Release Notes

DesignWare Building Blocks — Product Code: 2925-0

W-2024.09-SP1 Release for Synthesis

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Release Notes: Building Block IP

This document contains the latest information about the Synopsys® DesignWare® Building Block IP DWBB_202409.1 Release for Design Compiler®, Fusion Compiler™, Synplify Pro®, and VCS® MX. This library of IP was previously called the DesignWare Foundation (DWF) Library. The older DWF nomenclature appears in licensing and file naming. This release is based on the W-2024.09-SP1 Synthesis release.

This document also contains release information for DesignWare Building Block IP in FPGA synthesis (Synplify® tools).

The following lists the topics in this document:

- [“Features and Changes in DWBB_202409.1 / W-2024.09-SP1”](#) on page 6
- [“Known Problems and Limitations”](#) on page 6
- [“Features and Changes in Previous Versions \(History\)”](#) on page 7
- [“Obsoleted IP for New Designs”](#) on page 14
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1.1 Features and Changes in DWBB_202409.1 / W-2024.09-SP1

This section lists additions and updates for the current release of DesignWare Building Block IP. This section also describes the release information and issues related to using DesignWare Building Block IP with Synopsys Synplify Premier and Synplify Premier with Design Planner tools.

This release contains the following updates. In the following table, links in the left column (if present) go to more information in the STARS on the Web (SotW) system.

Table 1-1 STARS Resolved in DWBB_202409.1 (W-2024.09-SP1)

Component	STAR ID	Type	Description
Documentation	5729515	Enhancement	Remove old information from Release Notes (this document)
DW_ecc DW_lp_piped_ecc	5730254	Bug	Simulation model does not match synthesis for some 3-bit, 5-bit, 7-bit, ... errors
DW_lod DW_lsd DW_lza DW_lzd	5735087	Enhancement	Remove include file for function in simulation model
DW_pl_reg	5783512	Enhancement	Old synthesis coding style can mislead VC-SpyGlass detection of register reset type

1.2 Known Problems and Limitations

This section lists known problems and limitations. There are no known problems to report at this time.

1.3 Features and Changes in Previous Versions (History)

1.3.1 DWBB_202409.0 / W-2024.09

This release contains the following updates. In the following table, links in the left column (if present) go to more information in the STARS on the Web (SotW) system.

Table 1-2 STARS Resolved in DWBB_202409.0 (W-2024.09)

Component	STAR ID	Type	Description
Library	4863154	E	For tool compatibility, Verilog functions and tasks should be declared "automatic"

1.3.2 DWBB_202312.5 / V-2023.12-SP5

This release contains the following updates. In the following table, links in the left column (if present) go to more information in the STARS on the Web (SotW) system.

Table 1-3 STARS Resolved in DWBB_202312.5 (V-2023.12-SP5)

Component	STAR ID	Type	Description
DW_lp_fifoctl_1c_df	4452839	Bug	The RAM inside the Verilog sim model does not support the full depth range
DW_fp_recip DW_fp_recip_DG DW_fp_square	4712937	Bug	TINY flag is not behaving as the interpretation of the IEEE 754 standard states
DW_tap	4877359	Enhancement	DW_tap clock_dr and update_dr are tied to 0 after synthesis
DW_tap_uc	5014911	Enhancement	DW_tap_uc clock_dr and update_dr are tied to 0 after synthesis
DW_fp_addsub	5513269	Bug	Unexpected hierarchy ungrouping during logic_opto stage
DW01_prienc	5615272	Enhancement	Synthesizing DW01_prienc shows error of UID-342

1.3.3 DWBB_202312.4 / V-2023.12-SP4

This release contains the following updates. In the following table, links in the left column (if present) go to more information in the STARS on the Web (SotW) system.

Table 1-4 STARS Resolved in DWBB_202312.4 (V-2023.12-SP4)

Component	STAR ID	Type	Description
DW_lzd	4904185	Enhancement	Fix the placement of include file in DW_lzd
DW01_ash	5128655	Bug	Warning message "NULL Range found"

Table 1-4 STARs Resolved in DWBB_202312.4 (V-2023.12-SP4) (Continued)

Component	STAR ID	Type	Description
DW03_lfsr_dcnto DW03_lfsr_load DW03_lfsr_scnto DW03_lfsr_updn DW_crc_p DW_crc_s	5144382	Enhancement	Re-code low power simulation methodology to allow for third-party simulators
DW_lp_multifunc DW_lp_multifunc_DG	5389269	Bug	Incorrect statement in datasheet of DW_lp_multifunc(_DG)
DW_lp_fifo_1c_df	5415517	Bug	Synthesis tools cannot synthesize DW_lp_fifo_1c_df with synchronous reset

1.3.4 DWBB_202312.2 / V-2023.12-SP2

This release contains the following updates. In the following table, links in the left column (if present) go to more information in the STARs on the Web (SoTW) system.

Table 1-5 STARs Resolved in DWBB_202312.2 (V-2023.12-SP2)

Component	STAR ID	Type	Description
DW_div_seq DW_fp_div_seq DW_mult_seq DW_sqrt_seq	5094048	Enhancement	Use case-equality operator in to avoid an error message in VC S -xprop
DW_div_seq	5144148	Enhancement	DW_div_seq to gate an internal register for power improvement
DW_asymfifo_s2_sf	5261445	Bug	DW_asym_fifo_s2_sf fails synthesis when "depth" is 256
DW_fp_exp	5316771	Enhancement	Bad link to the Floating Point Overview in the datasheet for DW_fp_exp
DW_div_seq	5355510	Enhancement	Improve the code style of a register in the VHDL simulation model of DW_div_seq
DW_mult_seq	5355518	Bug	Functionality error if START is issued on the clock cycle when a previous operation is about to complete

1.3.5 DWBB_202312.1 / V-2023.12-SP1

This release contains the following updates. In the following table, links in the left column (if present) go to more information in the STARs on the Web (SotW) system.

Table 1-6 STARs Resolved in DWBB_202312.1 (V-2023.12-SP1)

Component	STAR ID	Type	Description
DW02_tree DW02_multp DW_squarep	4312369	Enhancement	Add "verif_en" value constraint checking to the VHDL simulation model
DW_ecc	5144145	Enhancement	Enable third-party simulator to run UPF simulation in Verilog
DW_fp_sum4	5144654	Bug	DW_fp_sum4 is not correctly handling some special cases that require higher internal precision

1.3.6 DWBB_202312.0 / V-2023.12

This release contains the following updates. In the following table, links in the left column (if present) go to more information in the STARs on the Web (SotW) system.

Table 1-7 STARs Resolved in DWBB_202312.0 (V-2023.12)

Component	STAR ID	Type	Description
DW_fp_dp3	4919691	Bug	Incorrect output in the synthesis model for DW_fp_dp3
DW02_multp DW02_tree DW_squarep	5057742	Enhancement	Allow non-VCS simulators to run the simulation model
DW_data_qsync_hl DW_lp_fifoctl_1c_df DW_lp_fp_multifunc DW_lp_fp_multifunc_DG DW_lp_multifunc DW_lp_multifunc_DG	5072702	Enhancement	Remove the use of 'ifndef SYNTHESIS' in the simulation model

1.3.7 DWBB_202212.5 / U-2022.12-SP5

This release contains the following updates. In the following table, links in the left column (if present) go to more information in the STARs on the Web (SotW) system.

Table 1-8 STARs Resolved in DWBB_202212.5 (U-2022.12-SP5)

Component	STAR ID	Type	Description
DW_In	9001562707	Bug	DW_In ulp errors are larger than the documented err_range The documented values for err_range have been corrected
DW_lp_piped_div DW_lp_piped_ecc DW_lp_piped_fp_add DW_lp_piped_fp_sum3 DW_lp_piped_mult DW_lp_piped_prod_sum DW_lp_piped_sqrt	4819539	Bug	VHDL simulation model of DW_lp_fp_piped_add returns uninitialized output
DW_arb_rr	4833770	Enhancement	In the datasheet, clarify the how the enable signal works

1.3.8 DWBB_202212.4 / U-2022.12-SP4

This release contains the following updates. In the following table, links in the left column (if present) go to more information in the STARs on the Web (SotW) system.

Table 1-9 STARs Resolved in DWBB_202212.4 (U-2022.12.SP4)

Component	STAR ID	Type	Description
DW_log2	9001562788	Bug	DW_log2 ulp errors are larger than err_range
DW_exp2	9001562658	Bug	DW_exp2 ulp errors are larger than err_range
DW_fp_div DW_fp_div_DG DW_fp_div_seq DW_lp_piped_fp_div	3638931	Bug	TINY flag is not behaving as the interpretation of the IEEE 754 standard states
DW_reset_sync	4351988	Enhancement	Update DW_reset_sync block diagram in datasheet
DW01_binenc DW01_prienc DW_lza DW_pricod DW_shifter	4760257	Enhancement	_function.inc generates errors in third-party simulation tool

1.3.9 DWBB_202212.3 / U-2022.12-SP3

The DesignWare Building Blocks IP library was not updated for this release.

1.3.10 DWBB_202212.2 / U-2022.12-SP2

This release contains the following updates. In the following table, links in the left column (if present) go to more information in the STARS on the Web (SotW) system.

Table 1-10 STARS Resolved in DWBB_202212.2 (U-2022.12-SP2)

Component	STAR ID	Type	Description
DW02_multp DW02_tree DW_squarep	4360627	Enhancement	Remove the carry-save randomness warnings from the warnings that are suppressed with DW_SUPPRESS_WARN

1.3.11 DWBB_202212.1 / U-2022.12-SP1

This release contains the following updates. In the following table, links in the left column (if present) go to more information in the STARS on the Web (SotW) system.

Table 1-11 STARS Resolved in DWBB_202212.1 (U-2022.12-SP1)

Component	STAR ID	Type	Description
DW_ecc	4479241	Enhancement	Add embedded script to disable ungrouping
DW_ecc DW_lp_piped_ecc	4548964	Bug	For some configurations, the simulation model is unable to detect and correct 1-bit errors in the most significant bit

1.3.12 DWBB_202212.0 / U-2022.12

This release contains the following updates. In the following table, links in the left column (if present) go to more information in the STARS on the Web (SotW) system.

Table 1-12 STARS Resolved in DWBB_202212.0 (U-2022.12)

Component	STAR ID	Type	Description
DW_div	4487770	Bug	The datasheet is missing the 'lut' synthesis implementation

1.3.13 DWBB_202203.5 / T-2022.03-SP5

This release contains the following updates. In the following table, links in the left column (if present) go to more information in the STARS on the Web (SoTW) system.

Table 1-13 STARS Resolved in DWBB_202203.5 (T-2022.03-SP5)

Component	STAR ID	Type	Description
DW_fp_exp	3310285	Bug	The datasheet shows incorrect maximum ulp error
DW_fp_exp2	3310283	Bug	The datasheet shows incorrect maximum ulp error
DW_fp_log2	3315120	Bug	The datasheet shows incorrect maximum ulp error

1.3.14 DWBB_202203.4 / T-2022.03-SP4

This release contains the following updates. In the following table, links in the left column (if present) go to more information in the STARS on the Web (StoW) system.

Table 1-14 STARS Resolved in DWBB_202203.4 (T-2022.03-SP4)

Component	STAR ID	Type	Description
DW_crc_p	3922446	Bug	Synthesis fatal for certain configurations
DW_log2 DW_fp_log2	4320062	Bug	Numerical error is larger than 2 ulp when <code>op_width >= 39</code>
DW_log2 DW_fp_log2	4337513	Enhancement	Improve the Verilog simulation model when <code>op_width >= 39</code>
DW_iir_sc	9001520839	Bug	Fusion Compiler fails for DW_iir_sc when negative parameter values are used

1.3.15 DWBB_202203.3 / T-2022.03-SP3

This release contains the following updates. In the following table, links in the left column (if present) go to more information in the STARS on the Web (StoW) system.

Table 1-15 STARS Resolved in DWBB_202203.3 (T-2022.03-SP3)

Component	STAR ID	Type	Description
DW_asymfifocltl_2c_df	3463163	Bug	Synthesis error when parameter <code>verif_en</code> is set to 2 or 3
DW_fifo_2c_df DW_fifocltl_2c_df	3850715	Enhancement	Enhance source domain output flags during requested clearing sequences
Library	3973672	Enhancement	Improve efficiency of the <code>dw_analyze_hybrid</code> script

Table 1-15 STARs Resolved in DWBB_202203.3 (T-2022.03-SP3) (Continued)

Component	STAR ID	Type	Description
DW_squarep DW02_multp DW02_tree	4306649	Enhancement	In the datasheet, clarify that the <i>verif_en</i> parameter affects only the simulation model and not the synthesis model

1.3.16 DWBB_202203.2 / T-2022.03-SP2

This release contains the following updates. In the following table, links in the left column (if present) go to more information in the STARs on the Web (StoW) system.

Table 1-16 STARs Resolved in DWBB_202203.2 (T-2022.03-SP2)

Component	STAR ID	Type	Description
DW01_csa	3825567	Enhancement	Enhance datasheet for usage of HDL operator
DW_ecc	3851321	Enhancement	Clarify error detection/correction behavior in datasheet
DW_fp_mult DW_fp_mult_DG	4175244	Enhancement	Add a new parameter (<i>en_ubr_flag</i>) that enables a status flag (<i>status[6]</i>) for “underflow before rounding” conditions
DW02_multp	4219789	Bug	Correct error in Table 1-5 of the datasheet for the <i>out1</i> signal when <i>verif_en</i> = 3

1.3.17 DWBB_202203.1 / T-2022.03-SP1

This release contains the following updates. In the following table, links in the left column (if present) go to more information in the STARs on the Web (StoW) system.

Table 1-17 STARs Resolved in DWBB_202203.1 (T-2022.03-SP1)

Component	STAR ID	Type	Description
DW_fp_div DW_fp_div_DG	3947569	Bug	TINY status behaves incorrectly in 'str' implementation when <i>ieee_compliance</i> =0
DW_fp_div DW_fp_div_DG	3928902	Enhancement	Add a new parameter (<i>en_ubr_flag</i>) that enables a status flag (<i>status[6]</i>) for “underflow before rounding” conditions

1.4 Obsolete IP for New Designs

DesignWare Building Block IP components are occasionally removed from the library. The process of removing components occurs in three stages over a period of at least two years.

1.4.1 Stage 1: Notification of Obsolete IP

The first stage in obsoleting IP is a notification that the IP will not be supported in the future. You can still simulate and synthesize the IP, but a warning is issued during simulation.

There is currently no DesignWare Building Block IP at this stage.

1.4.2 Stage 2: Notification of Obsolete Simulation Models

The second stage of the IP obsolescence is to remove the simulation model from the library. This occurs approximately 1 year from the time of the announcement that a particular IP is being obsoleted for new designs. Simulation is not possible, but synthesis of old designs still is possible.

There is currently no DesignWare Building Block IP at this stage.

1.4.3 Stage 3: Notification of Obsolete Synthesis Models

The third and final stage of the IP obsolescence process is to remove the remaining synthesis model from the library. This occurs approximately 2 years from the time of the announcement that a particular IP is being obsoleted for new designs. Simulation and synthesis is not possible.

The following DesignWare Building Block IP is at stage 3 of obsolescence: **none**.

1.4.4 Notification of Obsolete Synthesis Implementations

There are no recent obsoleted synthesis implementations.

1.5 Installation

DesignWare Building Block IP is installed with your Fusion Compiler or Design Compiler synthesis release.

1.5.1 How Can I Tell Which DesignWare Building Block IP Version Is Installed?

One way to view which DesignWare Building Block IP library you currently have installed on your UNIX system is to issue the following command:

```
% cat $SYNOPTSYS/dw/version
```

The string that is returned looks something like this:

```
W-2024.09-DWBB_202409.1
```

The “W-2024.09” portion indicates the Synthesis base release. The “DWBB_202409.1” portion indicates the DesignWare Building Block IP release, which uses the YYYYMM.S format (where ‘S’ is the service pack number).

1.5.2 Version Label for Technical Support

For future reference, a file exists under \$SYNOPTSYS/dw called “version”, which contains the release version label. You need to know this label when contacting Synopsys technical support.

1.6 Documentation

1.6.1 DesignWare Building Block IP Datasheets and Manuals

PDF files containing the complete set of current DesignWare Building Block IP datasheets and manuals exist in the directory:

```
$SYNOPTSYS/dw/doc
```

The latest documentation is also available on the Synopsys website at the following location:

<https://www.synopsys.com/dw/buildingblock.php>

To search for any Synopsys IP product, you can use the following location:

<https://www.synopsys.com/dw/ipsearch.php>

In the results, use the “More Information” link to navigate to documents for the component.

1.6.2 Using DesignWare Building Block IP In Other Applications

For information about running designs that contain DesignWare Building Block IP in various applications, see the SolvNetPlus article: [Running Implementation and Verification on Designs that Contain DesignWare Building Blocks IP](#).

1.7 Synopsys Statement on Inclusivity and Diversity

Synopsys is committed to creating an inclusive environment where every employee, customer, and partner feels welcomed. We are reviewing and removing exclusionary language from our products and supporting customer-facing collateral. Our effort also includes internal initiatives to remove biased language from our engineering and working environment, including terms that are embedded in our software and IPs. At the same time, we are working to ensure that our web content and software applications are usable to people of varying abilities. You may still find examples of non-inclusive language in our software or documentation as our IPs implement industry-standard specifications that are currently under review to remove exclusionary language.

1.8 Getting Help

1.8.1 Web Resources

- Fast access to DWBB component instantiation code snippets, function inference snippets (where appropriate), and links to documentation:
<http://www.synopsys.com/dw/buildingblock.php>
- To subscribe to receive proactive notifications on new product releases, information on Synopsys Technical Action Requests (STARs), product updates, and more (requires SolvNet login):
<https://www.synopsys.com/dw/mydesignware.php>
- Up-to-date information about the latest Synopsys IP:
<https://www.synopsys.com/designware-ip.html>
- General Synopsys Licensing (SCL) information:
<https://www.synopsys.com/keys>

1.8.2 Synopsys IP Component Subscriptions

For components you are entitled to, you can subscribe to receive notifications on new product releases, Synopsys Technical Action Request (STAR) reports, product updates and more at the following page:

<https://www.synopsys.com/dw/mydesignware.php>

1.8.3 STARs on the Web (SotW)

You must review all STARs on the Web (SotWs) associated with your product. SotWs are considered a part of the Synopsys documentation suite, and show critical information related to your product. To review product SotWs, refer to the Synopsys IP product information:

To see STAR reports, go to the following link and navigate to the desired component, which lists the link for the Open and/or Closed STARs reports:

<https://www.synopsys.com/dw/buildingblock.php>

1.8.4 Customer Support

Synopsys provides various methods for contacting Customer Support, as follows:

- For the fastest response, enter a case through SolvNetPlus:

a. <https://solvnetplus.synopsys.com>

**Note**

SolvNetPlus does not support Internet Explorer.

- b. Click the **Cases** menu and then click **Create a New Case** (below the list of cases).
- c. Complete the mandatory fields that are marked with an asterisk and click **Save**.
Make sure to include the following:
- **Product L1:** DesignWare Building Block IP
 - **Product L2:** Select the category of the component (such as Math_Arithmetic)
 - **Product L3:** Select the specific DesignWare Building Block IP component, if known
- d. After creating the case, attach any debug files you created.
- Or, send an e-mail message to support_center@synopsys.com (your email will be queued and then, on a first-come, first-served basis, manually routed to the correct support engineer):
 - Include the Product L1, Product L2, and Product L3 names, and Version number in your e-mail so it can be routed correctly.
 - For simulation issues, include the timestamp of any signals or locations in waveforms that are not understood
 - Attach any debug files you created.
 - Or, telephone your local support center:
 - North America:
Call 1-800-245-8005 from 7 AM to 5:30 PM Pacific time, Monday through Friday.
 - All other countries:
<https://www.synopsys.com/support/global-support-centers.html>

